

Mission Antyodaya 2020 Cluster Analysis

What 6 Lakh Indian Villages Tell Us About Rural Development

Amit Kumar, CoRE Stack | Research supervision: Prof. Aaditeshwar Seth, IIT Delhi

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The Mission Antyodaya 2020 survey records rural development through a large number of variables covering public services, institutions, infrastructure, livelihoods, agriculture, health, water, sanitation, and connectivity. Read separately, these variables are difficult to compare and do not readily show how different aspects of a village interact. Our analysis reduces this complexity to a smaller set of indices while retaining the component features needed for diagnosis and further analysis.

Data Preparation

Of the 198 parameters available across 32 state and union-territory files, 121 were selected for analysis. They were transformed into 64 features using normalization, binary availability measures, inversion, ratios, ordinal mappings, and composite scores. All features were oriented so that a higher value represents more of the measured positive condition. Examples include piped-water coverage, hours of domestic electricity, child nutrition and development, land utilization, agricultural risk support, and access to markets.

The 64 features were organized into 21 categories. Features were classified as Low, Medium, and High where sufficient variation existed; binary indicators used Low and High. Within each category, these classes were scored (Low = 0, Medium = 0.5, High = 1) and combined with equal weight. MiniBatch K-means was then used to classify the resulting category index. The published files retain the category result and the normalised feature values used to explain it. They do not publish feature-class columns.

Reading the Current Files

Begin with a `*_cat_cluster` field. It gives the final Low, Medium, or High class for one of the 21 categories. Next read the matching `*_cat_value`, then examine the normalised `*_feat_value` fields that contribute to it. These feature values remain comparative measures on a common 0–1 scale.

There is no `*_feat_cluster` field in the analytical CSV, GeoPackage, GeoServer layer, Excel sheet, or village Know Your Landscape output. The Excel and village report keep 21 category classes, 21 category values, and 109 raw survey fields. Raw fields are included only to trace a result back to the reported survey entries. They are not normalised or prepared for comparison, so they should not be used to rank villages or treated as analysis-ready indicators.

Interpreting the Results

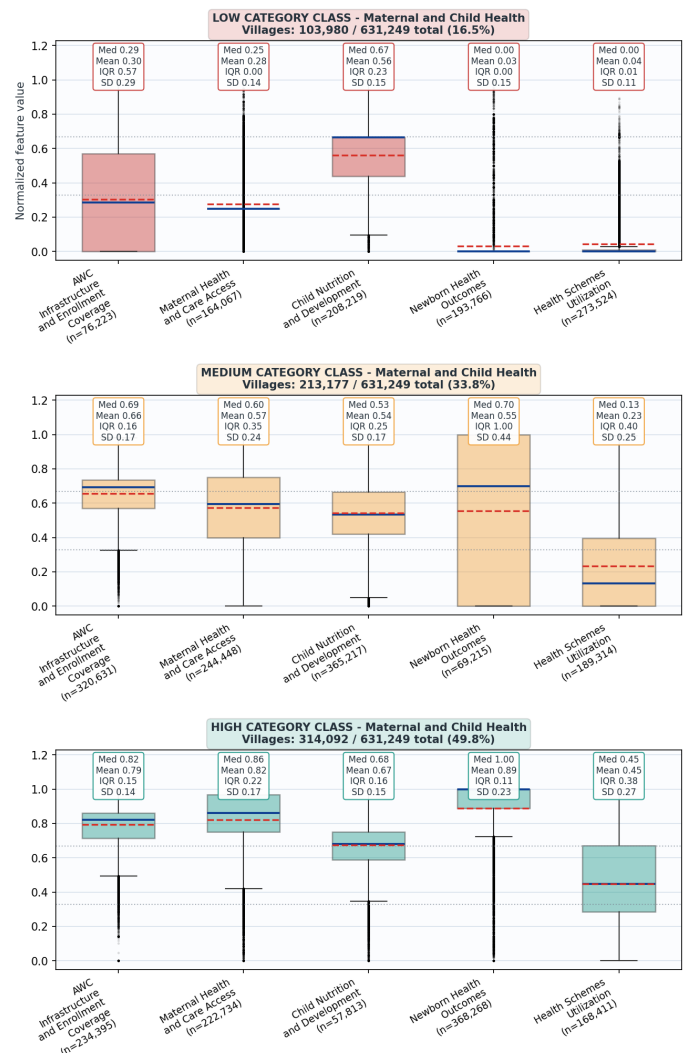
Figure 1 shows how to open a category and examine its components. Each panel contains villages in one final Maternal and Child Health category class: Low, Medium, or High. Within a panel, each box shows one of the five normalised feature values. The blue line is the median, the red dashed line is the mean, the box covers the middle half of observed values, and the whiskers and points show the wider spread. Reading this plot does not require a feature-class field.

Read the plots in two directions: follow one feature across the Low, Medium, and High panels, and then compare all features within one panel. This separates differences between classes

from internal gaps within a class.

The High class illustrates the point. Its median is 0.82 for Anganwadi infrastructure and enrolment, 0.86 for maternal health and care access, 0.68 for child nutrition and development, 1.00 for newborn health outcomes, and 0.45 for health-scheme utilization. Thus, “High” is a summary of the overall configuration; it does not mean that every component is uniformly high. The lower value and wider spread for health-scheme utilization identify an internal gap that would be hidden if only the category label were read.

Maternal and Child Health



Blue line = median | Red dashed line = mean | Box = Q1-Q3 | Whiskers = 1.5x IQR | n = villages in that feature class

Figure 1: Maternal and Child Health feature distributions across the three category classes.

Across classes, Anganwadi infrastructure and enrolment rises from a median of 0.29 in Low to 0.69 in Medium and 0.82 in High; maternal health and care access rises from 0.25 to

0.60 and 0.86; and newborn health outcomes rises from 0.00 to 0.70 and 1.00. The pattern is not uniformly monotonic: median Child Nutrition and Development is 0.67 in Low, 0.53 in Medium, and 0.68 in High. Health-scheme utilization improves across classes but remains comparatively weak even in High. Class membership therefore reflects the joint configuration of features, not a fixed threshold that every feature crosses.

The category label is a starting point. It should be read with component distributions, sample sizes, and feature definitions.

Using Category Patterns

Only the 21 category classes should be used to define patterns or filter villages. The category value and normalised feature values can then explain why a selected village falls within that pattern. For example:

Water-security pressure.

Pattern: High Livelihoods: Employment + Medium/High Agriculture: Land Cultivation + Low Agriculture: Irrigation and Watershed.

This combination points to places where livelihood dependence and cultivation are substantial but the measured water-security system is weak.

Agriculture without protective systems.

Pattern: Medium/High Agriculture: Land Cultivation + Low Agriculture: Support Services + Low Agricultural Markets + Low Financial Inclusion.

The filter brings together cultivation with weak support, finance, storage, and market access.

Institutions without economic conversion.

Pattern: High Institutionalization + Low Financial Inclusion + Low Agricultural Markets.

A stronger institutional base is present, but links to finance and markets remain weak.

Connectivity without local economic conversion: 34,931 villages.

Pattern: High Road Connectivity + High Energy Access + Low Financial Inclusion + Low Agricultural Markets + Low Cottage and Traditional Industry.

Roads and energy are present, while financial, market, and enterprise systems remain weak.

The same logic can identify opportunities. High Livelihoods: Common Resources with Low Livestock and Veterinary, or High Livelihoods: Fisheries with Low Financial Inclusion and Low Agricultural Markets, are category-level filters that can be checked against their normalised feature values and then verified locally.

Spatial Analysis

Figure 2 shows the next step after choosing the Water and Sanitation category. It maps the village-level `water_sanitation_cat_value` across India as a smooth display surface. Each hexagon summarises nearby mapped village values for legibility at national scale; it is not a district or state

average. The category class remains the filter, while this normalised value shows the strength and continuity of the spatial pattern.

Mission Antyodaya 2020: Village Water & Sanitation

Hexagonal display of 545,874 spatially linked village scores; each cell shows the mean of villages inside it.

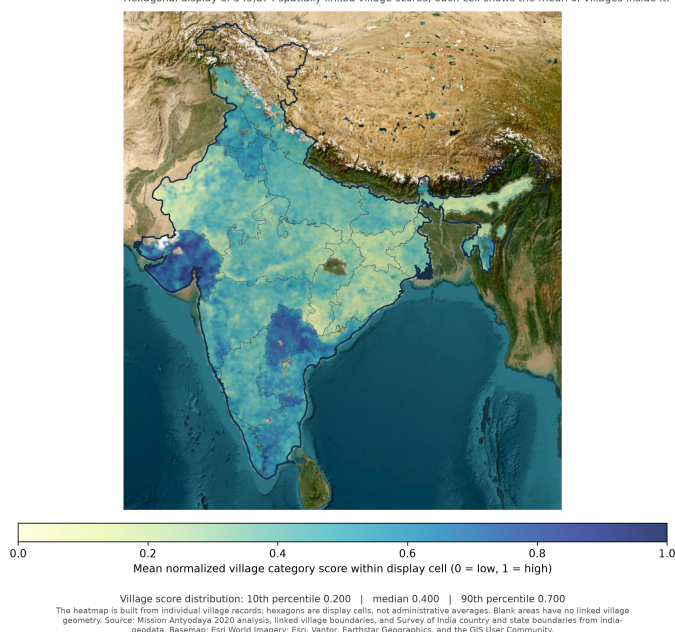


Figure 2: Village-derived Water and Sanitation category-value heatmap across India.

Users can return from the national surface to individual village category classes, compare the normalised category and feature values, check coverage, and finally trace the relevant raw survey entries. The same sequence works for all 21 categories.

Limitations

The classes are relative groupings in the 2020 data, not service standards, administrative rankings, causal findings, or programme recommendations. Missing values, uneven reporting, aggregation, ecology, and local economic history can affect interpretation. Results should be checked against feature definitions, current data, and field conditions. Raw survey fields can help trace a result, but cannot replace the normalised analytical values.

Conclusion

The value of the framework lies in moving between levels of evidence: from a category class, to its value, to the contributing normalised features, and finally to the raw survey entries that need checking. Used in this order, the analysis can narrow a field question without pretending that an unnormalised survey count is itself a comparable indicator.

References and data: Mission Antyodaya 2020 Village Cluster Analysis Report; village-level normalized values and cluster labels; pan-India GeoPackage; Mission Antyodaya 2020 raw data files; GEE Antyodaya 2020 Explorer. Data preparation and analytical assistance: Ananya Singh, IIT Delhi.